Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Intelligent Semiconductor Device Solution: Flexible Optical Frequency Comb**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (√), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical・Electronics		Electrical・Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Semiconductor Devices and Systems	Compound Semiconductor Devices	Optical Application Devices	Laser-related Components and Generation Devices
Concept of Proposed Technology	**Intelligent Semiconductor Device Solution: Flexible Optical Frequency Comb** ■ **Core Concept** AI semiconductor device solutions are becoming increasingly significant in the field of AI technology. This work aims to develop a high-performance, multi-channel optical frequency comb IC in a photonic integrated circuit (PIC) format for next-generation versatile semiconductor data processing, leveraging Fraunhofer HHI's proprietary optical modulator and RF IC platform technologies.			

	■ **Main Functions** ○ **High-Flatness Multi-channel Carrier Generation**: Generating dozens of precise, uniform optical channels (optical carriers) from a single laser input. ○ **Tunable Frequency Channel Spacing**: Precise control of the repetition rate and frequency spacing via RF signal modulation. ■ **Potential Uses** - Semiconductor Data Processing - Optical Interconnect - Dual Comb Spectroscopy - Optical Atomic Clocks - Microwave Generation - Timing and Synchronization
Rationale for Joint R&D Project	■ **Rationale for International Joint R&D Project** ○ **Leveraging World-Class Modulator Platforms for Innovative Outcomes**: By utilizing HHI's world-leading optical modulator and RF IC technology platforms, this project aims to produce groundbreaking research results that surpass the limitations of current optical signal processing. ○ **Securing Strategic Electro-Optic (EO) Optical Frequency Comb Technology**: EO-Comb technology is a vital core technology for future communications, radar, and spectroscopy. This collaboration is essential to secure proprietary designs and fabrication methodologies for high-performance, monolithically integrated EO-Combs.
Key Words	1. Optical Signal Processing 2. Optical Frequency Comb 3. Optical Modulator

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (4)	End of the Project	Level (7)

R&D Objectives

■ **Final Goal**

To develop an intelligent, monolithically integrated Electro-Optic (EO) Frequency Comb light source module for next-generation optical signal processing by leveraging high-performance modulator and RF IC platforms.

■ **Main Development Contents**

○ **Development of a compact, high-flatness multi-channel carrier source at the Photonic Integrated Circuit (PIC) level.**

Realizing a high-density integrated source optimized for next-generation optical interconnects and signal processing.

○ **Establishment of an industrial-grade optical engine platform to bridge the gap from laboratory-scale frequency combs.**

Transitioning complex benchtop EO-Comb systems into a robust, product-ready integrated module (TRL 7).

○ **Integration of intelligent control mechanisms for high-performance EO-Comb generation.**

Implementing advanced signal processing and control logic to ensure spectral stability and multi-channel consistency.

○ **Optimization of RF IC and Optical Modulator bonding for seamless signal integrity.**

Securing ultra-low-loss interfaces between high-speed RF components and the PIC platform to maximize modulation efficiency.

R&D Target Specifications

Performance Indicators	Unit	Target
Number of Channels	ea	≥20
Spectral Flatness	dB	≤3
Repetition rate	GHz	25
Modulation Bandwidth	GHz	

R&D Plan

■ **Key tasks**

○ **Analysis of EO-Comb Generation Mechanisms and Design Specification Finalization**

Conducting a detailed theoretical analysis of electro-optic comb generation and finalizing performance targets.

○ **RF IC Interface Design and Integration**

Developing high-frequency electrical interfaces to ensure signal integrity and stable modulation.

○ **Optical Modulator Design**

Architectural design of the modulator within the PIC to optimize bandwidth and efficiency.

○ **Optical Modulator Bonding and Assembly**

Implementation of advanced bonding techniques for mechanical and electrical stability of the integrated chip.

■ Methodologies

○ **Simulation-Driven Modeling**

Utilizing **Lumerical (FDTD/INTERCONNECT)** to model comb spectral flatness based on modulator bias conditions and RF driving power.

○ **Proprietary PIC Fabrication**

Applying the Fraunhofer HHI's **proprietary modulator process library** to fabricate chips with guaranteed high-speed response characteristics.

○ **High-Precision Packaging**

Developing ultra-precision alignment and bonding techniques to **minimize coupling loss** between the PIC chip and optical fibers.

○ **Performance Analysis and Characterization**

Verifying optical and electrical characteristics using high-resolution **Optical Spectrum Analyzers (OSA)** and **high-speed oscilloscopes**.

■ Milestone

Timeline	Major Deliverables
Month 6	**[Design Phase Completion]** Finalization of PIC Layout and Joint Roadmap Document
Month 18	**[Core Component Secured]** Acquisition of German-fabricated PIC Chips and Test Reports
Month 30	**[Prototype Integration]** Completion of Full Module Packaging and Target Spec Verification
Month 36	**[Project Finalization]** Submission of Field Validation Report and IP Strategy

■ Timeline

○ **Phase1 (M1-M6)**

✓ Simulation modeling, interface design, and requirement definition.

○ **Phase2 (M7-M36)**

✓ **Fabrication (M7-M18):** Foundry-based PIC manufacturing and unit testing.

	✓ **Packaging (M19-M30):** High-precision alignment, bonding, and module assembly-. ✓ **Validation (M31-M36):** Application-level testing and commercialization planning.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	■ **Technical Contributions** **Leveraging** Fraunhofer HHI's **world-class** high-speed modulator platform to contribute to the design of electro-optic modulation-based optical frequency comb generation structures and physical parameter optimization for multi-channel signal processing. ■ **Research Scope** ○ **PIC-based Frequency Comb Generation Simulation** Modeling of optical phenomena within Photonic Integrated Circuits (PIC) considering Fraunhofer HHI's foundry processes and deriving optimal driving conditions. ○ **Establishment of Packaging Guidelines** Analysis of optical coupling tolerances and definition of RF interface specifications to enhance packaging efficiency for Korean SME partners. ○ **System Integration and Demonstration Research** Performance verification of prototype modules in practical application environments, such as communication and signal processing. ■ **Resources to be Provided** ○ **Advanced Characterization Infrastructure** Support for utilizing specialized measurement equipment, including high-resolution Optical Spectrum Analyzers (OSA), ultra-high-speed oscilloscopes, and phase noise analyzers. ○ **High-Performance Modulator Platform and PIC Foundry** Access to Fraunhofer's **top-tier** high-speed/high-efficiency modulator process technology and Photonic Integrated Circuit (PIC) fabrication lines. ○ **Expert Personnel and Networks** Facilitating global technology trend sharing through the research institute's extensive international network. ■ **Specific Tasks**

Expected R&R of Korean SME Partner	1. **Deliverable**: Finalization of the EO-Comb generation layout and PIC design based on the HHI modulator platform. 2. **Support**: Drafting optical and electrical target specifications for prototype fabrication. 3. **Verification**: Precision characterization of multi-channel optical properties (≥ 20 channels, spectral flatness ≤3 dB) of packaged prototypes. 4. **Demonstration**: Performance demonstration of the multi-channel light source in practical signal processing and delivery of the final outcome report
	■ **Technical Contributions** ○ **Development of PIC-based Low-loss Optical Packaging Solutions** Contributing to ultra-precision alignment and coupling technologies between PIC chips fabricated at the Fraunhofer foundry and optical fibers/electrical signals. ○ **Optimization of RF Signal Integrity** Providing dedicated interface and wire-bonding technologies to ensure stable delivery of high-speed RF signals to the chip for continuous EO-Comb generation. ○ **Modular Thermal Management and Housing Design** Designing heat dissipation structures and compact module housings to ensure the long-term operational stability of the light source. ■ **Development Tasks** ○ **Fabrication of Integrated EO-Comb Module Prototypes** Manufacturing industrial-grade optical frequency comb module prototypes based on the research institute's design specifications. ○ **Packaging Process Optimization** Establishing standardized packaging processes, including chip bonding, optical alignment, and sealing, optimized for mass production. ○ **User Interface (UI) and Driving Circuit Development** Integrating power supply and control boards to facilitate stable operation and user-friendly control of the module. ■ **Testing/Validation Activities** ○ **Verification of Packaging Quality and Reliability** Performing precision measurements of post-packaging insertion loss (≤ 3 dB) and polarization-maintaining performance. ○ **Environmental Adaptability Testing**

Conducting operational stability and accelerated life tests under harsh conditions, including temperature fluctuations, vibration, and humidity.

○ **Field Applicability Evaluation**

Executing system interworking tests and preparing for standardization certification according to the requirements of final end-users.

■ Resource Commitments

○ **Allocation of Dedicated Production and Analytical Infrastructure**

Providing access to SME-owned specialized packaging facilities, such as cleanrooms, automatic optical alignment equipment, and reliability test chambers.

○ **Deployment of Expert Engineering Team**

Assigning a dedicated team of research personnel with extensive experience in optical component packaging and system integration to the project.

■ Collaboration Efforts

○ **Market Entry and Commercialization Initiatives**

Developing sales channels within domestic and international telecommunications equipment manufacturers and the data center market.

○ **Establishment of a Feedback System**

Collaborating on 'Design for Manufacturing (DfM)' enhancements by sharing packaging process issues and insights with the German research institute.

5. Expected Outcomes/Impact

Expected Outcomes and Impact

■ Technological Achievements

○ **Acquisition of Core EO-Comb Technology**

Securing world-class, high-flatness (≥ 20 channels) PIC-based frequency comb using the Fraunhofer HHI platform.

○ **System Miniaturization & Optimization**

Reducing footprint and power consumption by integrating complex benchtop systems into a single-chip module.

○ **TRL Elevation**

Advancing technology maturity from lab-scale validation (TRL 4) to an industrial-grade system prototype (TRL 7).

■ Commercialization Potential

○ **Targeting AI & 6G Markets**

Addressing the urgent demand for multi-channel light sources in 400 Gbps+ AI HPC servers and 6G infrastructure.

○ **Expansion into Precision Sensing**

Entering high-value spectroscopy and sensing markets by leveraging precise 25~50 GHz repetition rates.

○ **Cost Competitiveness**

Establishing a mass-production pipeline to replace expensive foreign optical modules with localized PIC solutions

■ Broader Industrial and Economic Impact

○ **Fostering New Industrial Ecosystems and Job Creation**

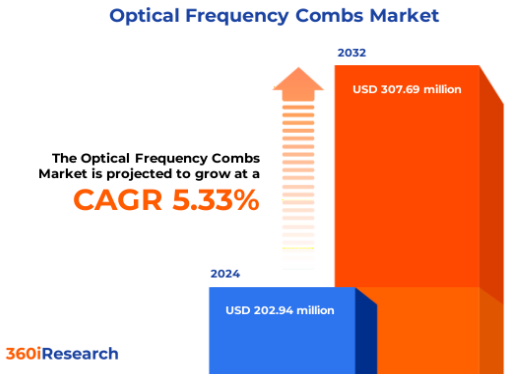
Driving growth in high-tech downstream industries such as AI processing, precision metrology, and quantum computing through the application of optical frequency comb, while generating high-value engineering jobs.

○ **Strengthening Global Technology Leadership**

Accelerating the development timeline through strategic collaboration with world-leading research institutions and expanding the global market share of Korean companies by securing international intellectual property (IP).

○ **Market Leadership**

Capitalizing on the growing optical frequency comb market, projected to reach USD 307.69 million by 2032 with a 5.33% CAGR.



6. References

(1) Bernal, S., Dumont, M., Berikaa, E. et al. 12.1 terabit/second data center interconnects using O-band coherent transmission with QD-MLL frequency combs. Nat Commun 15, 7741 (2024). https://doi.org/10.1038/s41467-024-51682-2

(2) Tao, Z., Wang, H., Feng, H. et al. Ultrabroadband on-chip photonics for full-spectrum wireless communications. Nature 645, 80–87 (2025). https://doi.org/10.1038/s41586-025-09451-8

(3) Fortier, Tara, and Esther Baumann. "20 years of developments in optical frequency comb technology and applications." *Communications Physics* 2.1 (2019): 153.

(4) Picqué, Nathalie, and Theodor W. Hänsch. "Frequency comb spectroscopy." *Nature Photonics* 13.3 (2019): 146-157.

(5) Millo, Jacques, et al. "Ultralow noise microwave generation with fiber-based optical frequency comb and application to atomic fountain clock." *Applied Physics Letters* 94.14 (2009).